

## Scalability & Future Plan

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|----------------------|-----------------------------------|
| <b>Date</b>          | 05 July 2026                      |
| <b>Team ID</b>       |                                   |
| <b>Project Name</b>  | Personalized Networking Assistant |
| <b>Maximum Marks</b> | 1 Mark                            |

### Scalability & Future Plan:

This document outlines the scalability and future enhancement plans for the Personalized Networking Assistant, including cloud deployment, LinkedIn integration, improved AI recommendations, enhanced security, and support for a larger user base and networking events.

### Current System Limitations:

| S.No | Limitation                                                          | Impact                                                           | Priority to Address (High / Medium / Low) |
|------|---------------------------------------------------------------------|------------------------------------------------------------------|-------------------------------------------|
| 1    | Uses limited user profile and event data for recommendations        | Reduces recommendation accuracy for diverse networking scenarios | High                                      |
| 2    | Application is deployed locally and cannot be accessed remotely     | Limits availability for users                                    | High                                      |
| 3    | Limited integration with external professional networking platforms | Reduces networking opportunities and profile enrichment          | Medium                                    |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                | Proposed Upgrade / Solution                                                                    |
|------|--------------------|------------------------------|------------------------------------------------------------------------------------------------|
| 1    | User Load          | Supports limited local users | Deploy on IBM Cloud, AWS, or Azure with scalable infrastructure                                |
| 2    | Data Storage       | SQLite database              | Upgrade to MySQL, PostgreSQL, or MongoDB for large-scale user data                             |
| 3    | Performance        | Local AI model inference     | Optimize DistilBERT and GPT-2 models, implement caching, and expose services through REST APIs |
| 4    | Security           | Basic user authentication    | Implement OAuth, HTTPS, and role-based access control, and encrypted user data                 |

**Future Roadmap:**

| Phase   | Planned Feature / Enhancement                                             | Target Timeline | Expected Impact                                                                                     |
|---------|---------------------------------------------------------------------------|-----------------|-----------------------------------------------------------------------------------------------------|
| Phase 1 | Integrate LinkedIn profiles and real-time event APIs                      | Within 2 Months | Provides richer user profiles and more relevant networking recommendations                          |
| Phase 2 | Add AI-based multilingual conversation generation and email notifications | Within 3 Months | Improves accessibility and enables users from different language backgrounds to network effectively |
| Phase 3 | Cloud Deployment and Mobile Application                                   | Within 6 Months | Improves accessibility and enables users from different language backgrounds to network effectively |